

Serum concentration of an inflammatory glycotoxin, methylglyoxal, is associated with increased cognitive decline in elderly individuals

Background Advanced glycations end products increase oxidant stress, inflammation, and neurotoxicity. Serum levels are increased in diabetes and aging. We examined the relationship between serum methylglyoxal derivatives (sMG), and cognitive decline, in 267 non-demented elderly. **Methods** Tobit mixed regression models assessed the association of baseline sMG with cognitive decline in the Mini Mental State Exam (MMSE) over time, controlling for sociodemographic factors (age, sex, and years of education), cardiovascular risk factors (diabetes and presence of an APOE4 allele), and kidney function. sMG was assessed by ELISA. **Results** The fully adjusted model showed an annual decline of 0.26 MMSE points per unit increase in baseline sMG ($p=0.03$). Significance was unchanged as additional risk factors were added to the model. The interactions of sMG with diabetes, sex, age, kidney function, and APOE4 genotype were not significant. **Conclusions** Higher levels of baseline sMG were associated with a faster rate of cognitive decline, after adjusting for several sociodemographic and clinical characteristics. This relationship did not differ by sex, APOE4 genotype, or diabetes status suggesting its generality. Since subjects were cognitively normal at the beginning of the study, elevated sMG may be indicative of brain cell injury initiated before clinically evident cognitive compromise.